Experiment 18

Matplotlib Library Use Cases in Data Science through Visualization.

Program:

```
import pandas as pd, numpy as np, seaborn as sns, matplotlib.pyplot as plt
%matplotlib inline
data=pd.read_csv("/content/Iris_Dataset.csv")
data.info(); data.describe(); data.value_counts("variety")
sns.countplot(x="variety",data=data); plt.show()
dummies=pd.get_dummies(data.variety)
FinalDataset=pd.concat([dummies,data.iloc[:,[0,1,2,3]]],axis=1)
FinalDataset.head()
sns.scatterplot(x="SepalLengthCm",y="SepalWidthCm",hue="variety",data=data)
sns.scatterplot(x="PetalLengthCm",y="PetalWidthCm",hue="variety",data=data)
sns.pairplot(data,hue="variety",height=3); plt.show()
for col in ["PetalLengthCm","PetalWidthCm","SepalLengthCm","SepalWidthCm"]:
    sns.FacetGrid(data,hue="variety",height=5).map(sns.histplot,col).add_legend()
    plt.show()
```

Output

